



First Name: _____

Family Name: _____

Student Number: _____

Diploma of Engineering
University Entrance Program
Foundation Studies Program

DPEN021 – Enabling Mathematics A

Skill Mastery Exam 2: Sample

This exam represents 24% of the total marks for the subject

Time allowed: 40 minutes
Items permitted in exam: UOW approved calculator

INSTRUCTIONS TO STUDENTS

1. Write in blue or black pens only.
2. All questions are to be attempted.
3. All notation is as given in lectures.
4. Show all necessary WORKING and reasoning in the spaces provided.
5. For full marks to be awarded, your working and reasoning need to be consistent with your selected answer. A correct answer with no working may receive a mark of zero.
6. You must achieve 80% on a Skill Mastery Exam 2 to be eligible to sit other exams in DPEN021.
7. Total number of questions: 12
 - All questions are multiple choice
 - Each question is worth 2 marks.
8. The answer to each question is to be written in the space provided.

No examination material may be removed from the examination venue.

UOW College A Registered Business of UOWC Limited
CRICOS Provider Code 02723D
A University of Wollongong Enterprise ABN 14105312329

Show all necessary WORKING and reasoning in the spaces provided. Use the space to do your working and then write your answer in the box provided.

Question 1 [Indices]

Simplify $2^{5x} \times 8^x \div 4^{2x}$

- (A) 2^{4x}
- (B) 2^{7x}
- (C) 2^{3x}
- (D) 2^{4x}
- (E) 2^x

Q1

Question 2 [Indices]

Simplify $\frac{-6a^{-1} \times (a^3b^{-1})^2}{3(a^{-4}b^3)^{-1}}$ and write your answer with positive indices.

- (A) $-2ab$
- (B) $\frac{-2b}{a}$
- (C) $\frac{-2a}{b}$
- (D) $-2a^9b$
- (E) $\frac{-2}{ab}$

Q2

Question 3 [Indices]

Simplify $\frac{\sqrt[6]{a^2b^3} \times \sqrt[3]{a}}{\sqrt{b}}$.

- (A) $a^{\frac{2}{3}}b$
- (B) $a^{\frac{2}{3}}$
- (C) b
- (D) $a^{\frac{3}{2}}b$
- (E) $a^{\frac{3}{2}}$

Q3

Question 4 [Solving Linear Equations]

Which statement is correct for this linear equation $\frac{5y+1}{4} = 6 - \frac{2y}{3}$?

- (A) $23y = 69$
- (B) $23y = 75$
- (C) $23y = 72$
- (D) $26y = 72$
- (E) $23y + 3 = 69$

Q4

Question 5 [Quadratic Equations-Quadratic Formula]

Using the quadratic formula, the solutions to $x^2 - 2x = \frac{1}{2}$ are:

(A) $x = \frac{2 \pm \sqrt{2}}{2}$

(B) $x = \frac{-2 \pm \sqrt{2}}{2}$

(C) $x = \frac{2 \pm \sqrt{6}}{2}$

(D) $x = \frac{1}{2}, 2\frac{1}{2}$

(E) $x = -\frac{1}{2}, \frac{3}{2}$

Q5

Question 6 [Quadratic Equations - Completing the Square]

Find the solution of the quadratic equation $3x^2 - 9x = 18$, using the method of completing the square. Leave your answer in exact form.

(A) $x = \frac{3 \pm \sqrt{33}}{2}$

(B) $x = \frac{-3 \pm \sqrt{33}}{2}$

(C) $x = \frac{3 \pm \sqrt{15}}{2}$

(D) $x = \frac{3 \pm \sqrt{9}}{2}$

(E) $x = \frac{3 \pm \sqrt{24}}{2}$

Q6

Question 7 [Reducible Quadratics]

Solve the equation $\left(x - \frac{5}{x}\right)^2 + 4\left(x - \frac{5}{x}\right) - 21 = 0$

- (A) $x = 3, x = -7$
(B) $x = -7, x = 3$
(C) $x = \frac{-7 \pm \sqrt{49 + 20}}{2} \quad x = \frac{-3 \pm \sqrt{9 + 20}}{2}$
(D) $x = \frac{-7 \pm \sqrt{49 + 20}}{2} \quad x = \frac{3 \pm \sqrt{9 + 20}}{2}$
(E) No solution

Q7

Question 8 [Solving Equations with Powers]

Solve the equation $2a^{\frac{2}{5}} - 2 = 6$ for a .

- (A) ± 32
(B) 32
(C) 4
(D) ± 2
(E) ± 8

Q8

Question 9 [Changing the Subject of an Equation]

Rearrange $\sqrt{\frac{1}{x} - \frac{1}{y}} = 3$ to make x the subject.

- (A) $x = \frac{y}{9y+1}$
- (B) $x = \frac{y}{3y+1}$
- (C) $x = \frac{y}{6y+1}$
- (D) $x = 9 + \frac{1}{y}$
- (E) $x = \frac{y}{9y-1}$

Q9

Question 10 [Inequality]

The values of x that satisfy the inequality $-9 \leq 2x - 1 < 9$, are:

- (A) $0 \leq x < 1$,
- (B) $-4 \leq x < 5$,
- (C) $\frac{1}{2} \leq x < \frac{7}{2}$,
- (D) $-\frac{1}{2} \leq x < \frac{1}{2}$,
- (E) no values of x exist.

Q10

Question 11 [Absolute Value Inequality]

Solve the inequality $|x - 3| < 2$.

- (A) $[1, 5]$
- (B) $(-\infty, \infty)$
- (C) $(-\infty, 1)$ or $(5, \infty)$
- (D) $(1, 5)$
- (E) $(-\infty, 1]$ or $[5, \infty)$

Q11

Question 12: [Absolute Value Equation]

Solve the absolute value equation $|x + 2| = 2x + 1$.

- (A) $x = 1$
- (B) $x = 1$ and $x = -1$
- (C) $x = -1$
- (D) $x = -2$
- (E) No solution

Q12

END OF EXAM

SAMPLE

Formula Sheet

Indices

Indices Rules:

$$a^n \times a^m = a^{n+m}$$

$$a^n \div a^m = a^{n-m}$$

$$(ab)^n = a^n b^n$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(a^n)^m = a^{n \times m}$$

$$a^0 = 1$$

$$a^{-n} = \frac{1}{a^n}$$

$$\frac{1}{a^{-n}} = a^n$$

$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$$

$$a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$$

Equations

Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Absolute Value Equations with Inequalities:

$$|x| < a \Rightarrow -a < x < a$$

$$|x| > a \Rightarrow x < -a \text{ or } x > a$$